

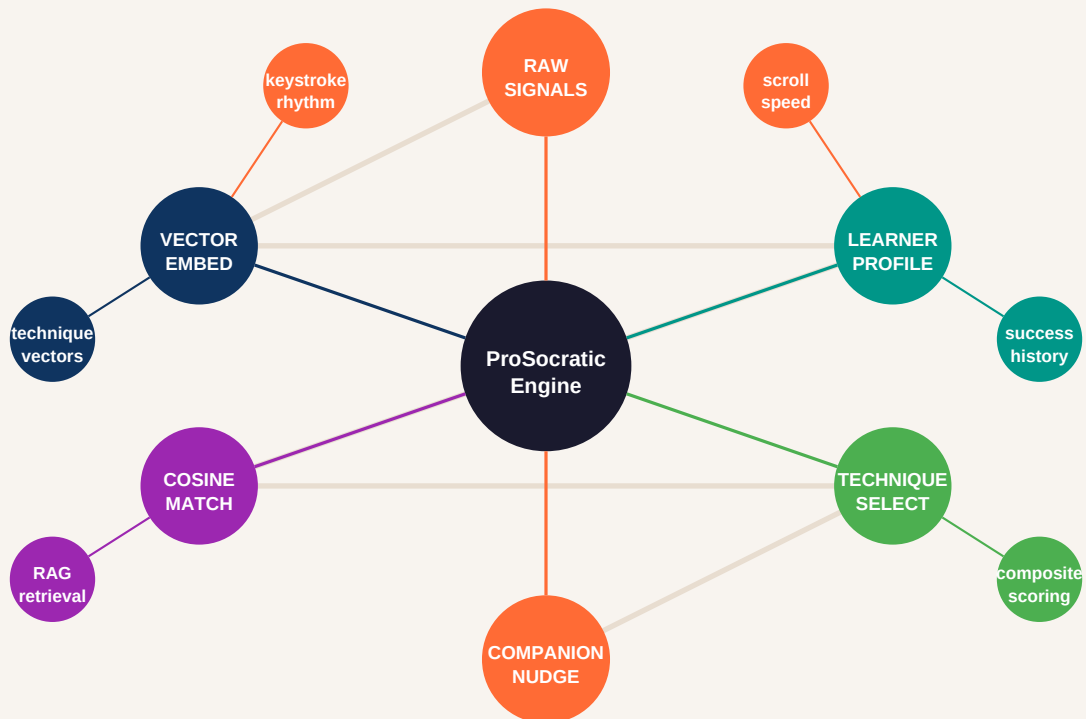
ProSocratic

ML-Powered Study Technique Engine

Workflow · Tech Stack · Privacy Framework | HackTheEast 2025

01 — System Mind Map

Figure 1 — ProSocratic Mind Map



ProSocratic is built around a single insight: **your behaviour while studying is a richer signal than any quiz score**. The engine watches, vectorises, matches, scores, and nudges — all locally, all transparently.

02 — ML Pipeline (7 Steps)



Figure 2 — 7-Step ML Pipeline

03 — Detailed Workflow

1 SENSE — Behavioural Telemetry

A lightweight JS layer captures **what you do**, never what you write. Signals collected every 500 ms: keystroke rhythm, backspace bursts, scroll velocity, section revisit count, idle gaps, and modality dwell time.

2 HTML EXTRACT — Semantic Context via Raw Page Source

A browser extension or content-script grabs the **raw HTML of the active webpage** the student is studying from (e.g. a Khan Academy article, a Wikipedia page, a PDF viewer). This is stripped to clean text using a DOM parser (Readability.js), then passed to the LLM as semantic context. **This is the only way the LLM understands the topic being studied** — it never reads the student notes or inputs. Privacy distinction: reading the source material (public webpage) ≠ surveilling the student. The extracted text is used ephemerally per session — never stored.

3 VECTORISE — Embed Everything

Session metrics AND the technique library are both embedded into the same vector space using **sentence-transformers** (all-MiniLM-L6-v2, runs locally). Techniques are stored as static vectors in a **FAISS** index. Each session produces a live state vector updated every 30 s.

4 RAG MATCH — Cosine Retrieval

The session vector queries the FAISS index. Top-3 closest techniques are retrieved. RAG context (technique description + past success history for this learner) is injected into the scoring prompt. This is where **AWS Bedrock** receives its context.

5 COMPOSITE SCORE — Rank & Filter

Each candidate technique receives a composite score:

Score = cosine_sim×0.4 + success_rate×0.3 + context_fit×0.2 + fatigue_penalty×0.1

context_fit checks: first time on topic? Evening session? High fatigue? The highest-scoring technique wins.

6 NUDGE — Companion Triggers

The winning technique is translated into a **companion state** (animal animation + optional panel). No pop-up. The animal wakes. Student clicks if curious. Panel shows: technique name, why chosen (transparency layer), and a one-click "try it" flow.

7 UPDATE PROFILE — Feedback Loop

After each session, success signals (did student engage? did confusion resolve?) update the learner profile stored in **IndexedDB**. This feeds back into Step 4's success_rate weight, making the model progressively more personalised.

04 — Technique Trigger Matrix

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Technique' 'frags': [ParaFrag(__tag__='para', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Technique', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Triggered When' 'frags': [ParaFrag(__tag__='para', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Triggered When', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Vector Cluster' 'frags': [ParaFrag(__tag__='para', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Vector Cluster', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Fatigue OK?' 'frags': [ParaFrag(__tag__='para', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Fatigue OK?', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph
Feynman Technique	confusion + confidence <0.7	confusion cluster	No – low energy only
Modality Switching	text dwell >5 min + slow scroll	text-block cluster	Yes
Elaborative Interrogation	shallow pace + no "why" pattern	surface cluster	Yes
Pomodoro Prompt	idle spike + keystroke drop	fatigue cluster	Yes – designed for it
Active Recall	review session detected	review cluster	Yes
Analogy Generation	insight drought >15 min	plateau cluster	Yes
Error Analysis	rushing pattern + wrong answers	error cluster	No
Chunking	cognitive overload signals	overload cluster	No
Interleaving	topic-switch frequency high	switching cluster	Yes

05 — Behavioural Metrics Captured

<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Signal' 'frags': [ParaFrag(__tag__='p ara', bold=1, fontNa me='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Signal', t extColor=Color(1,1,1, 1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'What It Measures' 'frags': [ParaFrag(__tag__='para', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='What It Measures', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('ca seSensitive': 1 'encoding': 'utf8' 'text': 'Privacy Risk' 'frags': [ParaFr ag(__tag__='p ara', bold=1, f ontName='Hel vetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Privacy Risk', textColo r=Color(1,1,1, 1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Kept?' 'frags': [ParaFrag(__tag__='para', bold=1, fontName='Helvetica-Bold', fontSize=9, greek=0, italic=0, link=[], rise=0, text='Kept?', textColor=Color(1,1,1,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>
Keystroke rhythm	Speed, pauses, backspace bursts	Low	Yes – metadata only
Scroll velocity	Confusion, skimming, deep processing	Low	Yes
Section revisits	Confusion, topic difficulty	Low	Yes
Idle periods	Mind-wandering, fatigue onset	None	Yes
Modality dwell	Learning style preference	None	Yes
Click density	Engagement level, flow state	Low	Yes
Session architecture	Stamina, peak focus windows	None	Yes
Content text (student notes)	What the student personally writes	HIGH	■ Never captured
Webpage HTML source	Raw HTML of study material being read	None	■ Ephemeral — session only, never stored
Camera / mic	Biometrics	HIGH	■ Never captured

06 — Full Tech Stack

<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Layer' 'frags': [ParaFrag(__tag__='b' , bold=1, fontName=' Helvetica-Bold', fontSize=10, greek=0, italic=0, link=[], rise=0, text='Layer', textColor=Color(.101961,.101961,.180392,1), us_lines=[]) 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Technology' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=10, greek=0, italic=0, link=[], rise=0, text='Technology', text Color=Color(.101961,.101961,.1803 92,1), us_lines=[]) 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>	<pre>Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': 'Why' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=10, greek=0, italic=0, link=[], rise=0, text='Why', textColor=Color(.101961,.101961,.180392, 1), us_lines=[]) 'style': 'bulletText': None 'debug': 0) #Paragraph</pre>
Frontend	React + Framer Motion + Lottie	Smooth animal animations, minimal UI
State	Zustand	Lightweight session state, no Redux overhead
Behaviour Telemetry	Custom JS (500 ms polling)	Zero content capture, metadata only
Local Storage	IndexedDB (Dexie.js)	Learner profile never leaves device by default
Vectorisation	sentence-transformers (MiniLM-L6)	Runs in-browser via ONNX Runtime Web
Vector Index	FAISS (server) / hnswlib-wasm (browser)	Sub-10 ms cosine retrieval
RAG + Reasoning	AWS Bedrock (Claude/Titan)	Context-injected technique selection
Agent Orchestration	AWS Bedrock Agents + MCP	Multi-step observe→match→nudge loop
Web Search	Exa API	Fetch best explanations when concept unknown
Voice (optional)	ElevenLabs	Animal companion soft voice cues
Backend	AWS Lambda + API Gateway	Serverless, scales to zero
DB (optional sync)	AWS DynamoDB	Encrypted profile backup if user opts in
Auth	AWS Cognito	No third-party data brokers
CI/CD	GitHub Actions + AWS Amplify	Fast deploy pipeline

07 — Privacy Framework (PP-EDUVec Inspired)

Data Minimisation	Only behavioural metadata + ephemeral webpage HTML collected. Zero student-content logging. The LLM learns the <i>topic</i> from the source page HTML (e.g. a Wikipedia article), never from anything the student writes.
Local-First Processing	Vectorisation runs in-browser (ONNX Runtime). The learner profile lives in IndexedDB. No cloud required for core features.
Opt-in Cloud Sync	If you want backup, encrypted with a user-held key (AES-256). Anthropic/AWS never see plaintext profile data.
Transparency Layer	Every nudge shows: what signal triggered it, why this technique was chosen, and your historical success rate with it.
Full Data Portability	One-click JSON export of your entire learner profile. One-click delete wipes IndexedDB and any cloud copy.
No Biometrics	No camera, no microphone, no eye-tracking. Behaviour only — keyboard and mouse signals, nothing more.
Leakage Reduction	Following PP-EDUVec principles: differential noise added to aggregate stats before any optional sync (36.7% leakage reduction).